# Recurrent Neural Networks and Language Models

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## 1 Language models

Informally, a language model is a function that, given a sequence of words, returns a probability distribution over the vocabulary: a probability for each word that could come next. It can also score a whole sequence, such as a sentence, with a single probability. Assigning a joint probability to a sequence of discrete tokens is useful in many domains besides language. However, natural language processing has shaped sequence modeling so strongly that the name "language model" is commonly used for such models, regardless of the kind of data they handle.

Applying the chain rule of probability turns language modeling into a series of next-token predictions. The joint probability of a sequence factors, from left to right, into conditional probabilities:

$$P(x_1, \dots, x_T) = P(x_1) \prod_{t=2}^{T} P(x_t \mid x_1, \dots, x_{t-1}).$$

With discrete tokens such as words, each factor is the output of a probabilistic classifier: the model reads the preceding tokens and returns a probability for every token in the vocabulary as the next one.

In practice, a model can condition on only a bounded number of preceding tokens, called its *context window*. Section 2 makes this informal idea precise.

## 2 Formal definition

Let $V$ be a finite vocabulary and let $K \geq 1$ be an integer, the *context window size*. For $n \geq 0$, $V^n$ denotes the set of sequences of length $n$ over $V$, with $V^0 = \{\varepsilon\}$, and $V^{<K} = \bigcup_{n=0}^{K-1} V^n$. For $x = (x_1, \dots, x_n)$ we write $x_{<t} = (x_1, \dots, x_{t-1})$, with $x_{<1} = \varepsilon$.

**Definition 1** (Language model with context window $K$)**.** A *language model with context window* $K$ is a probability distribution $P_K$ over $V^K$, i.e. a function $P_K : V^K \to [0, 1]$ with $\sum_{x \in V^K} P_K(x) = 1$.

**Definition 2** (Autoregressive parameterization)**.** An *autoregressive model with context window* $K$ is a family of conditional distributions $p(\cdot \mid c)$ over $V$, one for each context $c \in V^{<K}$. It defines

$$P_K(x_{1:K}) = \prod_{t=1}^{K} p(x_t \mid x_{<t}), \qquad x_{1:K} \in V^K.$$

Summing out $x_K, x_{K-1}, \dots, x_1$ in turn, each sum equals 1 because each $p(\cdot \mid c)$ is a distribution over $V$; hence $P_K$ is a language model with context window $K$. Each prediction conditions on at most $K - 1$ preceding tokens.